

# PAST EXAMINATION QUESTIONS (FP2)

## TOPIC 13: INTEGRATION (2) – REDUCTION FORMULAE

1. The integral  $I_n$ , where  $n$  is a positive integer, is defined by

$$I_n = \int_{\frac{1}{2}}^1 x^{-n} \sin \pi x \, dx.$$

- (i) Show that

$$n(n+1)I_{n+2} = 2^{n+1}n + \pi - \pi^2 I_n. \quad [5]$$

- (ii) Find  $I_5$  in terms of  $\pi$  and  $I_1$ .  $\left(I_5 = 4 + \frac{1}{24}(2\pi - 4\pi^2 - \pi^3 + \pi^4 I_1)\right) \quad [2]$

Oct/Nov 2019/11/12/13/3

2. It is given that, for  $n \geq 0$ ,

$$I_n = \int_0^1 x^n e^{x^3} \, dx.$$

- (i) Show that  $I_2 = \frac{1}{3}(e - 1)$ . [2]

- (ii) Show that, for  $n \geq 3$ ,

$$3I_n = e - (n-2)I_{n-3}. \quad [3]$$

- (iii) Hence find the exact value of  $I_8$ .  $\left(\frac{1}{3}(e - 2)\right) \quad [3]$

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3. Let  $I_n = \int_{\frac{1}{4}\pi}^{\frac{1}{2}\pi} \cot^n x \, dx$ , where  $n \geq 0$ .

- (i) By considering  $\frac{d}{dx}(\cot^{n+1} x)$ , or otherwise, show that

$$I_{n+2} = \frac{1}{n+1} - I_n. \quad [5]$$

May/June 2019/13/10 (part)

4. The curve  $C$  has equation

$$x^2 + 2xy = y^3 - 2.$$

- (i) Show that  $A(-1,1)$  is the only point on  $C$  with  $x$ -coordinate equal to  $-1$ . [2]

For  $n \geq 1$ , let  $A_n$  denote the value of  $\frac{d^n y}{dx^n}$  at the point  $A(-1,1)$ .

- (ii) Show that  $A_1 = 0$ . [3]

- (iii) Show that  $A_2 = \frac{2}{5}$ . [3]

Let

$$I_n = \int_{-1}^0 x^n \frac{d^n y}{dx^n} \, dx.$$

- (iv) Show that for  $n \geq 2$ ,

$$I_n = (-1)^{n+1} A_{n+1} - n I_{n-1}. \quad [3]$$

- (v) Deduce the value of  $I_3$  in terms of  $I_1$ . ( $I_3 = 0.4 + 6I_1$ ) [2]

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5. Let  $I_n = \int_1^{\sqrt{2}} (x^2 - 1)^n dx$ .

- (i) Show that, for  $n \geq 1$ ,

$$(2n + 1)I_n = \sqrt{2} - 2nI_{n-1}. \quad [5]$$

- (ii) Using the substitution  $x = \sec \theta$ , show that

$$I_n = \int_0^{\frac{1}{4}\pi} \tan^{2n+1} \theta \sec \theta d\theta. \quad [4]$$

- (iii) Deduce the exact value of

$$\int_0^{\frac{1}{4}\pi} \frac{\sin^7 \theta}{\cos^8 \theta} d\theta. \quad [5]$$

$$\left( \frac{16 - 9\sqrt{2}}{35} \right)$$

Oct/Nov 2018/12/11O

6. (i) Using the substitution  $u = \tan x$ , or otherwise, find  $\int \sec^2 x \tan^2 x dx$ . [2]  
It is given that, for  $n \geq 0$ ,

$$I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \tan^2 x dx. \quad \left( \frac{1}{3} \tan^3 x + c \right)$$

- (ii) Using the result that  $\frac{d}{dx}(\sec x) = \tan x \sec x$ , show that, for  $n \geq 2$ ,

$$(n + 1)I_n = (\sqrt{2})^{n-2} + (n - 2)I_{n-2}. \quad [5]$$

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7. (i) Show that

$$\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} e^x \cos x dx = \frac{1}{2} \left( e^{\frac{1}{2}\pi} + e^{-\frac{1}{2}\pi} \right). \quad [4]$$

- (ii) It is given that, for  $n \geq 0$ ,

$$I_n = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} e^{2x} \cos^n x dx.$$

Show that, for  $n \geq 2$ ,

$$4I_n = n(n-1) \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} e^{2x} \cos^{n-2} x \, dx - nI_n.$$

and deduce the reduction formula

$$(n^2 + 4n)I_n = n(n-1)I_{n-2}. \quad [6]$$

**May/June 2018/13/11E**

8. Let  $I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \, dx$  for  $n > 0$ .

- (i) Find the value of  $I_2$ . **(1)** [2]  
 (ii) Show that, for  $n \geq 2$ ,

$$(n-1)I_n = 2^{\frac{1}{2}n-1} + (n-2)I_{n-2}. \quad [5]$$

- (iii) The curve  $C$  has equation  $y = \sec^3 x$  for  $0 \leq x \leq \frac{1}{4}\pi$ . The region  $R$  is bounded by  $C$ , the  $x$ -axis, the  $y$ -axis and the line  $x = \frac{1}{4}\pi$ . Find the volume of revolution generated when  $R$  is rotated through  $2\pi$  radians about the  $x$ -axis.  $\left(\frac{28\pi}{15}\right)$  [4]

**Oct/Nov 2017/11/12/13/8**

9. Let  $I_n = \int_0^{\frac{1}{2}\pi} x^n \sin x \, dx$ .

- (i) Prove that, for  $n \geq 2$ ,

$$I_n + n(n-1)I_{n-2} = n\left(\frac{1}{2}\pi\right)^{n-1}. \quad [4]$$

- (ii) Calculate the exact value of  $I_1$  and deduce the exact value of  $I_3$ .  $\left(1; \frac{3}{4}\pi^2 - 6\right)$  [3]

**May/June 2017/11/12/6**

10. Let  $I_n$  denote  $\int_0^2 (4+x^2)^{-n} dx$ .

- (i) Find  $\frac{d}{dx}(x(4+x^2)^{-n})$  and hence show that

$$8nI_{n+1} = (2n-1)I_n + 2 \times 8^{-n}. \quad [5]$$

- (ii) Use the result for integrating  $\frac{1}{x^2+a^2}$  with respect to  $x$  in the List of Formulae (MF10), to find the value of  $I_1$  and deduce that

$$I_3 = \frac{3}{1024}\pi + \frac{1}{128}. \quad [5]$$

$$\left(I_1 = \frac{\pi}{8}\right)$$

**May/June 2017/13/6**

11. Evaluate  $\int_0^{\frac{1}{2}\pi} x \sin x \, dx$ . [2]

Given that  $I_n = \int_0^{\frac{1}{2}\pi} x^n \sin x \, dx$ , prove that, for  $n > 1$ ,

$$I_n = n \left( \frac{1}{2} \pi \right)^{n-1} - n(n-1)I_{n-2}. \quad [4]$$

By first using the substitution  $x = \cos^{-1} u$ , find the value of

$$\int_0^1 (\cos^{-1} u)^3 du,$$

giving your answer in an exact form.  $\left( \frac{3}{4} \pi^2 - 6 \right)$  [5]

**Oct/Nov 2016/11/12/13/9**

12. Let  $I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \sin^2 x \, dx$ , for  $n \geq 0$ . By differentiating  $\cos^{n-1} x \sin^3 x$  with respect to  $x$ , prove that

$$(n+2)I_n = (n-1)I_{n-2} \text{ for } n \geq 2. \quad [5]$$

Hence find the exact value of  $I_4$ .  $\left( I_4 = \frac{\pi}{32} \right)$  [4]

**May/June 2016/11/12/5**

13. Let  $I_n = \int_0^2 x^n (4-x^2)^{\frac{1}{2}} dx$ , for  $n \geq 1$ . By considering  $\frac{d}{dx} \left\{ x^n (4-x^2)^{\frac{3}{2}} \right\}$ , show that

$$(n+3)I_{n+1} = 4nI_{n-1}, \text{ where } n \geq 2. \quad [4]$$

Find the value of  $I_1$  and deduce the exact value of  $I_3$ .  $\left( I_1 = \frac{8}{3}, I_3 = \frac{64}{15} \right)$  [4]

**May/June 2016/13/6**

14. It is given that  $I_n = \int_1^e (\ln x)^n dx$  for  $n \geq 0$ . Show that

$$I_n = (n-1)\{I_{n-2} - I_{n-1}\} \text{ for } n \geq 2. \quad [6]$$

**Oct/Nov 2015/11/12/13/9**

15. Let  $I_n = \int_0^{\frac{1}{2}\pi} x^n \sin x \, dx$ , where  $n$  is a non-negative integer. Show that

$$I_n = n \left( \frac{1}{2} \pi \right)^{n-1} - n(n-1)I_{n-2}, \text{ for } n \geq 2. \quad [5]$$

Find the exact value of  $I_4$ .  $\left( \frac{1}{2} \pi^3 - 12\pi + 24 \right)$  [4]

**May/June 2015/11/12/7**

16. Let  $I_n = \int_0^{\frac{1}{2}\pi} \frac{\sin 2n\theta}{\cos \theta} d\theta$ , where  $n$  is a non-negative integer.

- (i) Use the identity  $\sin P + \sin Q \equiv 2 \sin \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q)$  to show that

$$I_n + I_{n+1} = \frac{2}{2n-1}, \text{ for all positive integers } n. \quad [5]$$

- (ii) Find the exact value of  $\int_0^{\frac{1}{2}\pi} \frac{\sin 8\theta}{\cos \theta} d\theta. \left(-\frac{152}{15}\right) \quad [4]$

**May/June 2015/13/5**

17. Let  $I_n = \int_0^1 (1-x)^n e^x dx$ . Show that, for all positive integers  $n$ ,

$$I_n = nI_{n-1} - 1, \quad [3]$$

Find the exact value of  $I_4$ . **(24e - 65)** [4]

By considering the area of the region enclosed by the  $x$ -axis, the  $y$ -axis and the curve with equation  $y = (1-x)^4 e^x$  in the interval  $0 \leq x \leq 1$ , show that

$$\frac{65}{24} < e < \frac{11}{4}. \quad [3]$$

**Oct/Nov 2014/11/12/13/7**

18. It is given that  $I_n = \int_0^{\frac{1}{4}\pi} \frac{\sin^{2n} x}{\cos x} dx$ , where  $n \geq 0$ . Show that

$$I_n - I_{n+1} = \frac{2^{-(n+\frac{1}{2})}}{2n+1}, \quad [5]$$

Hence show that  $\int_0^{\frac{1}{4}\pi} \frac{\sin^6 x}{\cos x} dx = \ln(1 + \sqrt{2}) - \frac{73}{120}\sqrt{2}. \quad [3]$

**May/June 2014/11/12/10**

19. Using the substitution  $u = \cos \theta$ , or any other method, find  $\int \sin \theta \cos^2 \theta d\theta. \left(-\frac{1}{3} \cos^3 \theta + c\right) \quad [1]$

It is given that  $I_n = \int_0^{\frac{1}{2}\pi} \sin^n \theta \cos^2 \theta d\theta$ , for  $n \geq 0$ . Show that, for  $n \geq 2$ ,

$$I_n = \frac{n-1}{n+2} I_{n+2}. \quad [5]$$

Hence find the exact value of  $\int_0^{\frac{1}{2}\pi} \sin^4 \theta \cos^2 \theta d\theta. \left(\frac{\pi}{32}\right) \quad [4]$

**May/June 2014/13/9**

20. It is given that

$$I_n = \int_0^1 \frac{x^n}{\sqrt{1+2x}} dx.$$

Show that, for  $n \geq 1$ ,

[3]

$$(2n+1)I_n = \sqrt{3} - nI_{n-1}.$$

Show that

[4]

$$I_3 = \frac{2}{35}(\sqrt{3} + 1).$$

**Oct/Nov 2013/11/12/4**

21. Let  $I_n = \int_0^1 (1+x^2)^n dx$ . Show that, for all integers  $n$ ,

[5]

$$(2n+1)I_n = 2nI_{n-1} + 2^n.$$

Evaluate  $I_0$  and hence find  $I_3$ . ( $I_0 = 1, I_3 = \frac{96}{53}$ )

[4]

Given that  $I_{-1} = \frac{1}{4}\pi$ , find  $I_{-3}$ . ( $I_{-3} = \frac{3}{32}\pi + \frac{1}{4}$ )

[5]

**Oct/Nov 2013/13/11E**

22. Let  $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$ . Prove that, for every positive integer  $n$ ,

[5]

$$2nI_{n+1} = 2^{-n} + (2n-1)I_n.$$

Given that  $I_1 = \frac{1}{4}\pi$ , find the exact value of  $I_3$ . ( $\frac{1}{4} + \frac{3}{32}\pi$ )

[3]

**May/June 2013/11/12/4**

23. Show that  $\int_0^1 xe^{-x^2} dx = \frac{1}{2} - \frac{1}{2e}$ .

[2]

Let  $I_n = \int_0^1 x^n e^{-x^2} dx$ . Show that  $I_{2n+1} = nI_{2n-1} - \frac{1}{2e}$  for  $n \geq 1$ .

[3]

Find the exact value of  $I_7$ . ( $I_7 = 3 - \frac{8}{e}$ )

[3]

**May/June 2013/13/5**

24. (ii) Let  $I_n = \int_0^\alpha \tanh^{2n} \theta d\theta$  for  $n \geq 0$ , where  $\alpha > 0$ .

[4]

(a) Show that  $I_{n-1} - I_n = \frac{\tanh^{2n-1} \alpha}{2n-1}$  for  $n \geq 1$ .

Given that  $\alpha = \frac{1}{2} \ln 3$ ,

(b) evaluate  $I_0$ . ( $\frac{1}{2} \ln 3$ )

[1]

**CAIE 1348/May/June 2013/1/13 (part)**

25. Let  $I_n$  denote  $\int_0^\infty x^n e^{-2x} dx$ . Show that  $I_n = \frac{1}{2}nI_{n-1}$ , for  $n \geq 1$ . [2]

Oct/Nov 2012/11/12/5

26. Show that

$$\int_0^x e^x \sin x dx = \frac{1 + e^\pi}{2}. \quad [4]$$

Given that

$$I_n = \int_0^\pi e^x \sin^n x dx,$$

show that, for  $n \geq 2$ ,

$$I_n = n(n-1) \int_0^\pi e^x \cos^2 x \sin^{n-2} x dx - nI_n,$$

and deduce that

$$(n^2 + 1)I_n = n(n-1)I_{n-2}. \quad [6]$$

May/June 2012/11/12/11E

27. Let

$$I_n = \int_1^e x^2 (\ln x)^n dx,$$

for  $n \geq 0$ . Show that, for all  $n \geq 1$ ,

$$I_n = \frac{1}{3}e^3 - \frac{1}{3}nI_{n-1}. \quad [4]$$

Find the exact value of  $I_3$ .  $\left(I_3 = \frac{(4e^3+2)}{27}\right)$  [4]

May/June 2012/13/4

28. Let  $I_n = \int_0^1 x^n (1-x)^{\frac{1}{2}} dx$ , for  $n \geq 0$ . Show that, for  $n \geq 1$ ,

$$(3 + 2n)I_n = 2nI_{n-1}. \quad [5]$$

Hence find the exact value of  $I_3$ .  $\left(I_3 = \frac{32}{315}\right)$  [3]

Oct/Nov 2011/11/12/6

29. Show that  $\frac{d}{dt}(t(1+t^3)^n) = (3n+1)(1+t^3)^n - 3n(1+t^3)^{n-1}$ . [3]

Let  $I_n = \int_0^1 (1+t^3)^n dt$ . Using the above result, or otherwise, show that

$$(3n + 1)I_n = 2^n + 3nI_{n-1}. \quad [2]$$

Hence evaluate  $I_3$ .  $\left(\frac{319}{140}\right)$  [4]

**Oct/Nov 2011/13/7**

**30.** Let

$$I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \, dx,$$

where  $n \geq 0$ . Show that, for all  $n \geq 2$ ,

$$I_n = \frac{n-1}{n} I_{n-2}. \quad [4]$$

**May/June 2011/11/12/10**

**31.** Let

$$I_n = \int_0^{\frac{1}{4}\pi} \tan^n x \, dx,$$

where  $n \geq 0$ . Use the fact that  $\tan^2 x = \sec^2 x - 1$  to show that, for  $n \geq 2$ ,

$$I_n = \frac{1}{n-1} - I_{n-2}. \quad [4]$$

Show that  $I_8 = \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \frac{1}{4}\pi$ . [4]

**May/June 2011/13/5**

**32.** Let  $I_n = \int_0^1 (1-x)^n \sin x \, dx$  for  $n \geq 0$ . Show that

$$I_{n+2} = 1 - (n+1)(n+2)I_n. \quad [4]$$

Hence find the value of  $I_6$ , correct to 4 decimal places. **(0.0177)** [4]

**Oct/Nov 2010/1/5**

**33.** Let

$$I_n = \int_1^e x(\ln x)^n \, dx,$$

where  $n \geq 1$ . Show that

$$I_{n+1} = \frac{1}{2}e^2 - \frac{1}{2}(n+1)I_n. \quad [3]$$

**May/June 2010/11/12/5**



34. Let

$$I_n = \int_0^{\frac{1}{2}\pi} \sin^n \theta \, d\theta,$$

where  $n$  is a non-negative integer. Show that  $I_{n+2} = \frac{n+1}{n+2} I_n$ . [4]

May/June 2010/13/9

35. Show that

$$\frac{d}{dx} \left[ x^{n-1} \sqrt{4-x^2} \right] = \frac{4(n-1)x^{n-2}}{\sqrt{4-x^2}} - \frac{nx}{\sqrt{4-x^2}}. \quad [3]$$

Let

$$I_n = \int_0^1 \frac{x^n}{\sqrt{4-x^2}} \, dx,$$

where  $n \geq 0$ . Prove that

$$nI_n = 4(n-1)I_{n-2} - \sqrt{3},$$

for  $n \geq 2$ . [2]

Given that  $I_0 = \frac{1}{6}\pi$ , find  $I_4$ , leaving your answer in an exact form.  $\left(\pi - \frac{7\sqrt{3}}{4}\right)$  [4]

Oct/Nov 2009/1/6

36. Let

$$I_n = \int_0^1 t^n e^{-t} \, dt,$$

Where  $n \geq 0$ . Show that, for all  $n \geq 1$ ,

$$I_n = nI_{n-1} - e^{-1}. \quad [3]$$

May/June 2009/1/4

37. Let  $I_n = \int_0^1 \frac{1}{(1+x^4)^n} \, dx$ . By considering  $\frac{d}{dx} \left( \frac{x}{(1+x^4)^n} \right)$ , show that

$$4nI_{n+1} = \frac{1}{2^n} + (4n-1)I_n. \quad [4]$$

Given that  $I_1 = 0.86697$ , correct to 5 decimal places, find  $I_3$ . (0.710) [4]

Oct/Nov 2008/1/7

38. (i) Given that

$$I_n = \int_0^{\frac{1}{2}\pi} t^n \sin t \, dt,$$

show that, for  $n \geq 2$ ,

$$I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}. \quad [5]$$

**May/June 2008/1/8 (part)**

**39.** Let  $I_n = \int_0^{\ln 2} (e^x + e^{-x})^n dx$ .

(i) Show that

$$\frac{d}{dx} [(e^x - e^{-x})(e^x + e^{-x})^{n-1}] = n(e^x + e^{-x})^n - 4(n-1)(e^x + e^{-x})^{n-2}. \quad [3]$$

(ii) Hence show that

$$nI_n = 4(n-1)I_{n-2} + \frac{3}{2} \left(\frac{5}{2}\right)^{n-1}. \quad [2]$$

**Oct/Nov 2007/1/8**

**40.** Let

$$I_n = \int_0^1 \frac{1}{(4-x^2)^n} dx,$$

for  $n = 1, 2, 3, \dots$  Verify that

$$\frac{d}{dx} \left[ \frac{x}{(4-x^2)^n} \right] = \frac{1-2n}{(4-x^2)^n} + \frac{8n}{(4-x^2)^{n+1}}$$

and hence prove that

$$8nI_{n+1} = (2n-1)I_n + \frac{1}{3n}. \quad [5]$$

**May/June 2007/1/9**

**41.** The integral  $I_n$ , where  $n$  is a non-negative integer, is defined by

$$I_n = \int_0^1 x^n e^{-x^3} dx.$$

By considering  $\frac{d}{dx} (x^{n+1} e^{-x^3})$  or otherwise, show that

$$3I_{n+3} = (n+1)I_n - e^{-1}. \quad [3]$$

Hence find  $I_6$  in terms of  $e$  and  $I_0$ . ( $I_6 = \frac{4I_0 - 7e^{-1}}{9}$ ) [2]

Oct/Nov 2006/1/2

42. Given that

$$I_n = \int_0^{\frac{1}{2}\pi} \sin^n x \, dx,$$

where  $n \geq 0$ , prove that

$$I_{n+2} = \left(\frac{n+1}{n+2}\right) I_n. \quad [4]$$

May/June 2006/1/7

43. Let

$$I_n = \int_0^1 (1+x^2)^{-n} dx,$$

where  $n \geq 1$ . By considering  $\frac{d}{dx}(x(1+x^2)^{-n})$ , or otherwise, prove that

$$2nI_{n+1} = (2n-1)I_n + 2^{-n}. \quad [5]$$

Deduce that  $I_3 = \frac{3}{32}\pi + \frac{1}{4}$ . [3]

Oct/Nov 2005/1/6

44. The integral  $I_n$ , where  $n$  is a non-negative integer, is defined by

$$I_n = \int_0^1 e^{-x}(1-x)^n dx.$$

(i) Show that  $I_{n+1} = 1 - (n+1)I_n$ . [3]

May/June 2005/1/8 (part)

45. Let

$$I_n = \int_e^{e^2} (\ln x)^n dx,$$

where  $n \geq 0$ . By considering  $\frac{d}{dx}[x(\ln x)^{n+1}]$ , or otherwise, show that

$$I_{n+1} = 2^{n+1}e^2 - e - (n+1)I_n. \quad [4]$$

May/June 2004/1/6

46. The integral  $I_n$  is defined by

$$I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \, dx.$$

By considering  $\frac{d}{dx}(\tan x \sec^n x)$ , or otherwise, show that

$$(n+1)I_{n+2} = 2^{\frac{1}{2}n} + nI_n. \quad [4]$$

Find the value of  $I_6$ .  $\left(\frac{28}{15}\right)$  [4]

**Oct/Nov 2003/1/5**

47. It is given that, for  $n \geq 0$ ,

$$I_n = \int_0^1 x^n e^{-x^3} \, dx.$$

(i) Find  $I_1$  in terms of  $e$ .  $\left(\frac{e-1}{2e}\right)$  [1]

(ii) Show that

$$I_{n+2} = \frac{n+1}{2} I_n - \frac{1}{2e}. \quad [3]$$

(iii) Find  $I_5$  in terms of  $e$ .  $\left(\frac{2e-5}{2e}\right)$  [2]

**Oct/Nov 2002/1/4**

48. Given that

$$C_n = \int_0^1 (1-x)^n \cos x \, dx \quad \text{and} \quad S_n = \int_0^1 (1-x)^n \sin x \, dx,$$

show that, for  $n \geq 1$ ,

$$C_n = nS_{n-1} \quad \text{and} \quad S_n = 1 - nC_{n-1}. \quad [3]$$

Hence find the value of  $S_3$ , correct to 6 decimal places. **(0.048826)** [3]

**May/June 2002/1/2**

49. Show that

$$\frac{d}{dx} \left( \frac{x}{(1+x^2)^n} \right) = \frac{2n}{(1+x^2)^{n+1}} - \frac{2n-1}{(1+x^2)^n}. \quad [4]$$

Hence show that

$$\frac{d}{dx} \left( \frac{x \sinh^{-1} x}{(1+x^2)^n} \right) = \frac{2n \sinh^{-1} x}{(1+x^2)^{n+1}} - \frac{(2n-1) \sinh^{-1} x}{(1+x^2)^n} + \frac{x}{(1+x^2)^{n+\frac{1}{2}}}. \quad [3]$$

It is given that

$$I_n = \int_0^1 \frac{\sinh^{-1} x}{(1+x^2)^n} dx.$$

Show that, provided  $n \neq \frac{1}{2}$ ,

$$2nI_{n+1} = (2n-1)I_n + 2^{-n} \ln(1+\sqrt{2}) - \frac{1-2^{-n+\frac{1}{2}}}{2n-1}. \quad [5]$$

Deduce that

$$I_2 = \frac{1}{2}I_1 + \frac{1}{4} \ln(1+\sqrt{2}) - \frac{1}{4}(2-\sqrt{2}). \quad [2]$$

**Nov 1996/1/3**

**50. (a)** Given that

$$I_n = \int_0^{\ln 3} \tanh^{2n} x \, dx \quad (n \geq 0),$$

show that

$$I_n - I_{n+1} = \frac{1}{2n+1} \left(\frac{4}{5}\right)^{2n+1}. \quad [5]$$

**June 1995/1/3 (part)**

**51. (b)** Given that

$$I_n = \int_0^1 (1-x)^n \sinh x \, dx,$$

show that, for  $n \geq 2$ ,

$$I_n = -1 + n(n-1)I_{n-2}. \quad [5]$$

**June 1992/1/4 (part)**

**52. (b)** Let

$$I_n = \int_0^{\ln 2} \operatorname{sech}^n x \, dx.$$

(i) Show that

$$\frac{d}{dx} (\operatorname{sech}^{n-2} x \tanh x) = (n-1) \operatorname{sech}^n x - (n-2) \operatorname{sech}^{n-2} x. \quad [4]$$

(ii) Deduce that

$$(n-1)I_n = (n-2)I_{n-2} + \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^{n-2} \quad [3]$$

**December 1991/1/3 (part)**

53. Given that

$$I_n = \int_0^{\ln 3} \sinh^n x \, dx \quad (n \geq 0),$$

show that, for  $n \geq 2$ ,

$$nI_n = \left(\frac{5}{3}\right)\left(\frac{4}{3}\right)^{n-1} - (n-1)I_{n-2}. \quad [6]$$

Deduce, or show otherwise, that  $I_2 = \frac{10}{9} - \frac{1}{2} \ln 3$ , and obtain a corresponding expression for  $I_4$  in terms of  $\ln 3$ . [5]

$$\left(I_4 = \frac{25}{162} + \frac{3}{8} \ln 3\right)$$

**December 1989/1/4 (part)**

54. (b) Given that

$$I_n = \int_0^1 \cosh^n x \, dx,$$

show that

$$nI_n = ab^{n-1} + (n-1)I_{n-2}$$

where  $a = \sinh(1)$  and  $b = \cosh(1)$ .

Deduce that

$$I_5 = \frac{a}{15}(8 + 4b^2 + 3b^4). \quad [7]$$

**June 1987/1/3 (part)**

55. Given that  $I_n = \int \tanh^n x \, dx$ , show that, for  $n \geq 2$ ,

$$I_n = I_{n-2} - \frac{1}{n-1} \tanh^{n-1} x + \text{constant},$$

and find  $I_5$ .  $\left(\ln(\cosh x) - \frac{1}{2} \tanh^2 x - \frac{1}{4} \tanh^4 x + c\right)$  [7]

Find, correct to three significant figures, the volume obtained when the region enclosed by the curve  $y = \tanh^3 x$ , the  $x$ -axis and the line  $x = 1$  is rotated through  $2\pi$  about the  $x$ -axis. [5]

(0.125)

June 1984/1/17

56. Differentiate  $\frac{\sinh x}{\cosh^{n-1} x}$  with respect to  $x$ . Hence, or otherwise, given that

$$I_n = \int_0^t \frac{1}{\cosh^n x} dx,$$

Prove that

$$(n-1)I_n = \frac{\sinh t}{\cosh^{n-1} t} + (n-2)I_{n-2}.$$

State the exact numerical values of  $\sinh(\ln 2)$  and  $\cosh(\ln 2)$  and show that

$$\int_0^{\ln 2} \frac{1}{\cosh^4 x} dx = \frac{66}{125}.$$

June 1980/1/17